Are We Chasing Shadows? Quantifying Unattributable Polarization and Attributing the Rest to Annotator Groups

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Abstract

Polarization has recently been proposed as a robust way of distinguishing minor disagreements from systematic differences in opinion for imbalanced, minority groups but there currently exists no methodology for attributing polarization to these groups. In this study, we discover and investigate two major limitations in realistic settings: (1) the presence of "inherent" polarization that cannot be attributed to any known or latent groups, and (2) opposing polarization effects canceling each other out in aggregated annotations. To address these issues, we introduce a new metric that attributes polarization to any annotator group while avoiding these limitations, tailored to smaller datasets common in NLP, and provide an open-source Python library implementation. We apply our method to four content moderation datasets and confirm that gender and ethnicity consistently explain polarization patterns, while ideological attributes may be more important than previously thought. Our work indicates that subjective NLP datasets may need fewer items and more annotators per item, estimated to be less than 20, for reliable inference.

1 Introduction

Annotations are essential for subjective tasks, such as content moderation, toxicity, and Hate Speech (HS) detection, which are especially important for protecting minority groups (United Nations, 2025; Dioum, 2025). However, in practice, analysis rests on inter-annotator agreement or majority-vote (Ivey et al., 2025; Kralj Novak et al., 2022; van der Velden et al., 2025), which may lead to biased and misconfigured systems (e.g., toxicity models disproportionally marking African American (AA) speech as "toxic" (Sap et al., 2019)), or discard *fundamental* disagreements between minority groups and the majority (Quaranto, 2022), (see the example in Fig. 1).[1]

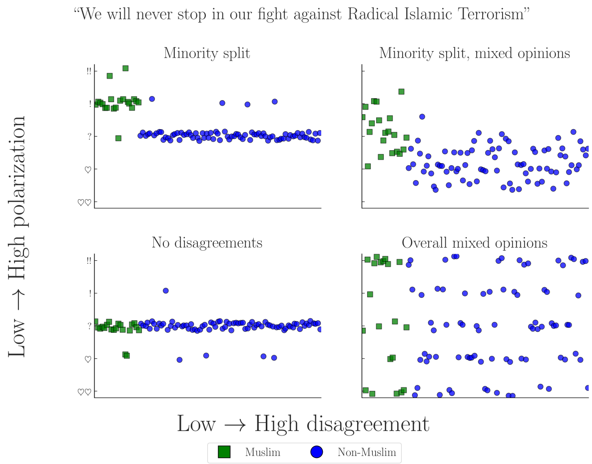

Figure 1: Simulated experiment exploring four scenarios of a Hate Speech (HS) detection task applied to a controversial statement by D. Trump (Jan. 2019). The variance in annotations across different annotators may cause traditional agreement metrics to ignore minority opinions, especially when opinions are mixed (top-right).

One alternative is using *polarization*, which correlates better with human judgments and is much less sensitive to different group sizes (Pavlopoulos and Likas, 2023, 2024). While we can detect polarization, there is little work on how to detect whether it is *actually caused by marginalized groups*. Our work thus tries to answer the following Research Question (RQ):

RQ. *How much polarization can we practically attribute to annotator subgroups given existing datasets?*

While prior work assumed that all polarization can be explained and attributed to some

[1]The simulated experiments in this study were implemented by generating random annotations following the normal distribution with differing means and Standard Deviation (SD)—see App. F.

annotator characteristic, through the exhaustive creation of theoretical annotator groups our study discovers the existence of Inherent Polarization (InPol), which can not be attributed to *any possible group* of annotators—known or unknown (Finding 1). We also discover that polarization across comments has a *direction*, preventing us from aggregating annotations on the dataset level to circumvent annotation sparsity (Finding 2).

In order to solve these issues we introduce a new metric—"*apunim*"—which attributes polarization to specific annotator sociodemographic and ideological characteristics (referred to as Personal Characteristics (PCs) in this study). Our metric is generalizable across any domain and modality (e.g., video/image classification) and, unlike previous approaches, enables both *quantitative* comparisons across datasets and statistical analysis via p-value estimation. Even in cases where unattributable polarization exists, *apunim* is capable of detecting general patterns of polarization across multiple groups. Additionally, our metric is designed to be used for existing Natural Language Processing (NLP) datasets which are usually small w.r.t. both content and annotations due to budget constraints (Chen et al., 2022; Golazizian et al., 2024).

We then apply our metric to four NLP datasets focused on Toxicity and HS detection (§6), finding that annotator gender and ethnicity significantly impact annotation (Finding 3). Our findings also indicate that ideological attributes may play a major role in explaining polarization (e.g., the more religious the annotators, the more polarized they become—Finding 4). Finally, we discuss the applicability of our methodology to the modern NLP dataset landscape (§7), and release an open source Python (PyPi) library that implements our metric.[2]

Warning: This publication includes examples of controversial and potentially harmful speech for the purposes of demonstration. These examples do not represent the views of the authors.

2 Background and related work

2.1 Agreement

Popular inter-annotator agreement metrics such as Cohen's kappa, Krippendorff's alpha, and Fleiss's kappa aim to distinguish genuine disagreement from random disagreement through statistical chance-correction. However, these metrics are often difficult to apply to intergroup agreement without extensions or restrictive assumptions (e.g., no missing labels or equal numbers of annotations). They have also been criticized for failing to account for differences arising from social, cultural, and demographic backgrounds (Feinstein and Cicchetti, 1990; Byrt et al., 1993; van der Velden et al., 2025; Checco et al., 2017), producing results that are difficult to compare across studies, being sensitive to class imbalance (Feinstein and Cicchetti, 1990), and relying on questionable assumptions about chance agreement (Checco et al., 2017). While recent work has addressed some of these limitations with methods that improve applicability and robustness (van der Velden et al., 2025; Wong et al., 2021; Ivey et al., 2025), they are much more complex and less intuitive compared to their traditional counterparts, indicating that agreement may not be the correct tool for analysis of minority viewpoints.

2.2 Polarization

There are multiple competing formulations for polarization. One of these approaches (Akhtar et al., 2019) utilizes χ^2 to estimate in-group vs out-group variance, but still requires traditional agreement metrics for analysis. Others attempt to solve this problem by framing disagreement as a cluster identification problem, applying either parametric (Checco et al., 2017), or non-parametric (Pavlopoulos and Likas, 2024; Mignemi et al., 2024) approaches to the annotation histograms.

For our work, we use a non-parametric polarization metric, normalized Distance From Unimodality (nDFU), which has been shown to correlate better with human judgment of disagreement than established metrics (Pavlopoulos and Likas, 2024). nDFU takes values in the range $[0, 1]$, where values close

[2]Library: https://anonymous.4open.science/r/apunim-5F70, Webpage: https://anonymous.4open.science/r/apunim-page, Replication code: https://anonymous.4open.science/r/aposteriori-unimodality-E8C1

to 0 indicate no polarization (i.e., opinions are concentrated around a single mode). The metric is defined as follows:

$$\mathrm{nDFU}(X) = \frac{\max_{i \neq m} \left(f_i - \begin{cases} f_{i-1}, & \text{if } i > m \\ f_{i+1}, & \text{if } i < m \end{cases} \right)}{f_m}$$
$$m = \arg\max_i f_i \tag{1}$$

where, X denotes the set of annotations, f_i the relative frequency of the i-th annotation (e.g., the frequency of 2 in a scale of 1–5), and m the histogram peak. Intuitively, nDFU measures how prominent secondary peaks are compared to the main peak (see Fig. 2 for an example).

2.3 Attributing polarization to groups

While many studies study the effect of subgroups on annotation agreement (Goyal et al., 2022; Binns et al., 2017; Giorgi et al., 2025; Al Kuwatly et al., 2020; Sachdeva et al., 2022), attributing *polarization* is a relatively new concept. Under the name *aposteriori unimodality* (Pavlopoulos and Likas, 2024), the original idea is that if polarization in a single item is positive, but zero for all annotator subgroups, then it is explained by those groups. Though promising, this mechanism operates on an edge-case scenario where *all* polarization is explained by a single split, and one that we disprove in this work (§4). Additionally, getting polarization estimates from few annotations per item is extremely noisy, which is especially problematic when the output is binary (i.e., an item either is or is not polarizing) instead of a quantifiable value.[3]

3 Datasets

We use four datasets from current NLP research that include PC information at the level of individual annotators; specifically, the "Kumar" (Kumar et al., 2021) and "Sap" ("Breadth of Posts") (Sap et al., 2022) datasets, which are concerned with Toxicity and HS detection respectively, and the DICES-350 and DICES-990 datasets (Aroyo et al., 2023), which focus on LLM response safety.[4]

Dataset	#Com	#Ann	#Dims	Task
Kumar	2998	5	10	Tox
Sap	585	5–6	3	HS
DICES-350	350	70–76	4	HS
DICES-990	990	123	4	HS

Table 1: Datasets used in this study. *#Ann* refers to the number of annotations per comment (see App. B), and *#Dims* to PC dimensions (e.g., age, gender). Tasks are either Toxicity (Tox) or HS.

A brief overview of dataset characteristics is provided in Table 1. The Kumar dataset stands out for its large size, featuring more than 100,000 annotated comments, as well as 10 distinct PC dimensions.[5] In contrast, the DICES datasets contain fewer annotated comments but involve more than an order of magnitude more annotators per comment than the other datasets. The Kumar and Sap datasets contain numerous polarizing items, unlike the DICES datasets, where most items are unpolarized (App. B; Fig. 7).[6] More details on the annotations can be found in App. B.

4 Is it possible to attribute all polarization to annotator groups?

4.1 Problem Formulation

Let $d = \{c_1, c_2, \ldots\}$ be a dataset composed of multiple annotated comments.[7] Each comment c is assigned multiple annotators, which creates the annotation multi-set $A(c) = \{(a_1, g_1), (a_2, g_2), \ldots\}$, where a_i is a single annotation (e.g., 2 in a LIKERT scale of 1–5), and g_i a group (e.g., men). As an example, the annotations for a comment in a sentiment analysis task where we group the annotators by gender, would be formulated as:

[3] For example, when presented with noisy estimates of a metric in the $[0, 1]$ range, we would ideally want to distinguish between cases where attribution is close to .001, and cases where attribution is .49.

[4] For the latter, we consider only labels relating to racism (`Q3_bias_overall`), thereby transforming the task to HS detection.

[5] We sample 3,000 comments in this study due to computational constraints (App. E). We run ablations for both different and differently sized-samples. Our results remain consistent, although large samples (10,000+) tend to produce null results (App. H).

[6] This discrepancy could be attributed to these datasets containing LLM-generated dialogues with models tuned to avoid unsafe speech, making it much easier to distinguish than in human comments.

[7] The annotated items could be of any modality, such images, videos or survey questions, even though we only consider comments in this study.

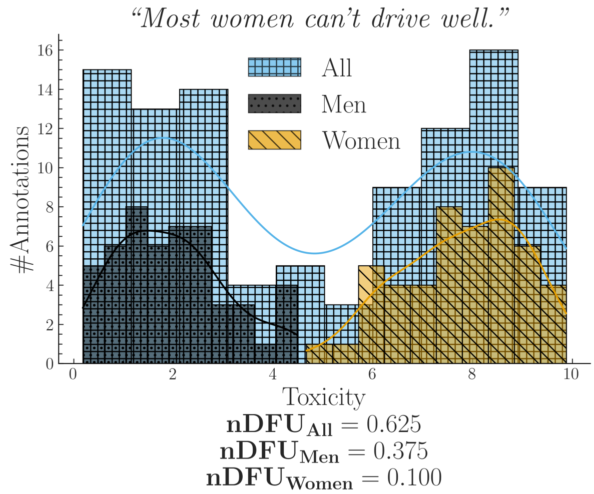

Figure 2: Simulated experiment describing a polarizing comment where male and female annotators agree between themselves, but disagree with the opposite gender.

$$A(\text{"Could be better, could be worse"}) = \{\ (+, F), (-, M), (+, F), \dots\ \}$$

Intuitively, a group partially explains the polarization of a comment c when the annotations of that group $A(c \mid g) = \{a(c, g') \in A(c) \mid g' = g\}$ exhibit higher polarization compared to the full set of annotations $A(c)$. Consider the case where a misogynistic comment is annotated for toxicity by male and female annotators shown in Fig. 2. The annotations are generally polarized ($nDFU_{all} = .62$); but both female ($nDFU_{women} = .10$) and male ($nDFU_{men} = .37$) annotators exhibit low polarization between their respective groups. This pattern suggests that the overall polarization is partially caused by substantive disagreements between male and female annotators.[8]

4.2 Inherent Polarization

4.2.1 What is it?

Inherent Polarization (InPol) refers to disagreement that cannot be attributed to *any specific* grouping of annotators; that is, polarization that exists no matter how we split the annotators in any arbitrary group. Prior work (Pavlopoulos and Likas, 2024) attributed polarization to annotator groups only when the observed polarization for each individual group became exactly zero (e.g. $nDFU(all) > nDFU(men) = nDFU(women) = 0$). The presence of InPol would invalidate this theory, since there would be no possible group or combination of groups g *s.t.* $nDFU(g) = 0$.

Definition 1 (Inherent Polarization)**.** Polarization that cannot be explained by any known or latent annotator grouping.

Formally, the InPol for a single comment c is the minimum polarization exhibited by any arbitrary group:

$$Pol_{inh}(c) = \min_{S \subseteq A(c),\, |S| \geq 3} \{nDFU(S)\} \qquad (2)$$

When considered exhaustively, these theoretical, random groups include both observed and latent groups (e.g., annotators may be less strict if they had lunch prior to annotation). Since nDFU operates only when a group has at least three annotators (Eq. 1), this yields $\sum_{k=3}^{n} \binom{n}{k}$ possible groups.

4.2.2 Experimental setup

We systematically explore the space of annotator groups by generating random partitions for each comment. Since enumerating all subsets is computationally intractable for large annotator counts, we establish an upper bound via Monte Carlo sampling by drawing k random partitions with random sizes:

$$Pol_{inh}^{MC}(c) = \min_{j=1}^{k} \{nDFU(\tilde{r}_j(A(c)))\} \qquad (3)$$

where $\tilde{r}$ is an operator that selects a randomly sized, randomly selected subset. Since Eq. 3 considers a strict subset of Eq. 2, it provides an upper bound for it:

$$Pol_{inh}^{MC}(c) \geq Pol_{inh}(c) \forall c \qquad (4)$$

4.2.3 Does it exist?

We compute the InPol for each comment across the four datasets. For Sap and Kumar, we use the exhaustive formulation (Eq. 2), whereas for the two DICES datasets we rely on the Monte Carlo approximation with $k = 1000$ (Eq. 3). Fig. 3 shows that in all DICES comments there exists at least one group that fully accounts for the observed polarization.[9] Interestingly, this observation does not hold for

[8] Since polarization persists among male annotators, another subgroup of men may also contribute to it.

[9] $Pol_{inh}^{MC}(c) = 0 \Rightarrow Pol_{inh}(c) = 0$ due to Eq. 4. The same logic applies if we tried to increase the number of random partitions in Eq. 3.

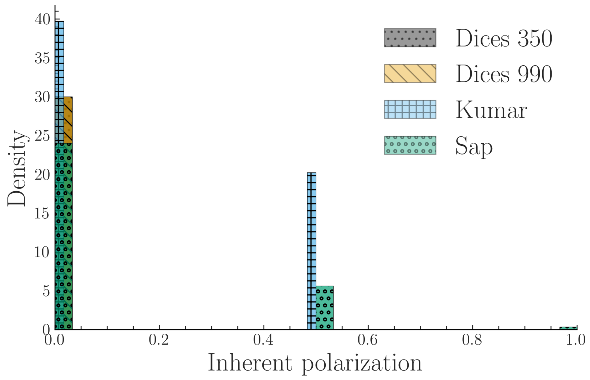

Figure 3: InPol across our datasets (§3). Contrary to assumptions made in prior work, InPol is commonly non-zero and can even reach $nDFU = 1$ in datasets with low annotator counts.

the Kumar and Sap datasets, despite evaluating all possible annotator groups exhaustively. See Table 9; App. G for some examples of comments featuring high InPol compared to those with none.

Finding 1. *Comments may exhibit a degree of InPol that cannot be attributed to any known or latent annotator grouping given the annotators present in the dataset.*

4.2.4 Why does it exist?

Hypothesis 1 (Low annotator count)**.** A low number of annotators per item may provide our model with insufficient information.

Hypothesis 2 (Inaccessible groups)**.** InPol may arise from the presence of distinct, partially represented groups within the annotator pool. While these minority groups are responsible for the observed polarization, they may go undetected in our current methodology if they are represented by only one or two annotators, as these groups are discarded (§4.2).

Both of these explanations are plausible and reflect known limitations of most NLP datasets, although we lack such datasets with large annotator counts per item to provide a definitive answer, since the DICES datasets are insufficient:

Why is DICES silent? Initially, we hypothesized that the absence of InPol in the DICES datasets could be attributed to their large annotator count per item. However, even when repeating the experiment using random annotator subsets (ranging from 5 to 50 annotators) and taking the maximum across ten random samples for each subset size, we still observed zero InPol (App. G). We believe that fundamental differences between human and LLM-generated texts may be the cause. This is a plausible explanation given that LLM texts violating anti-discrimination directives may be relatively unsubtle, as supported by low overall polarization (Fig. 7; App B).

4.3 Polarization is directional

One way of potentially overcoming annotator sparsity, is aggregating annotations between comments. Fig. 4 shows an experiment where we combine annotations across two comments. We observe that, should the two comments be polarized for the opposite reason, the opposing polarization effects might balance each other out, leading to a false negative. Therefore, any statistics must be computed on the comment-level; annotations should never be aggregated across comments.[10]

Finding 2. *Polarization has a unique direction for each comment which may compromise cross-comment aggregation.*

5 How do we attribute the rest?

5.1 Apriori polarization

Instead of using InPol, which represents the minimum possible polarization exhibited by any group, we calculate the *expected* (average). Therefore, we do not test whether the group is polarized against the whole; nor whether the group explains *some polarization* (as would be the case if we used Eq. 2), but rather whether the group explains polarization *more than other groups.*

Definition 2 (Apriori polarization)**.** The expected polarization of a comment's annotations.

Specifically, we define apriori polarization for a comment c as:

$$Pol_{apr}(c) = \frac{1}{t}\sum_{i=1}^{t} nDFU(\tilde{r}_i(A(c))) \quad (5)$$

where $\tilde{r}$ is the random partition operator (see Eq. 2). We explain why defining apriori polarization in this way is preferable to simpler alternatives in App. C.1.

[10]A similar assumption was followed in early parametric approaches, which assumed different parameters for each comment (Checco et al., 2017).

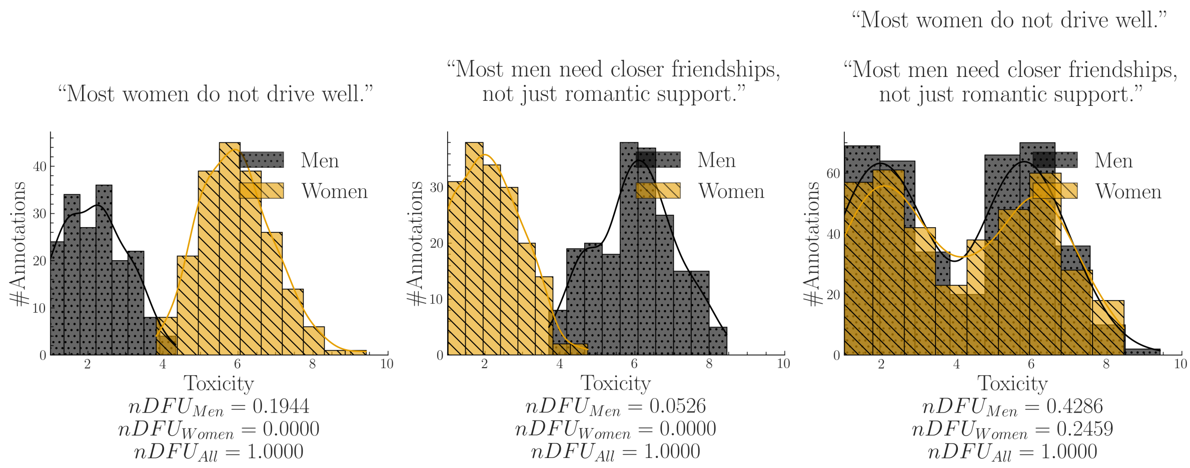

Figure 4: A synthetic experiment simulating a polarizing discussion with two comments where annotators disagree about which one is toxic. When we aggregate the comments, we unintentionally change the comparison: instead of comparing two unimodal distributions (each comment viewed separately) with two bimodal distributions (their aggregated annotations), we end up comparing two bimodal distributions, obscuring the source of the polarization (polarization explained by gender, but not shown).

5.2 Expanding to the whole dataset

We can estimate how much polarization is generally present in the dataset by simply averaging the *apriori* polarization of its comments. Intuitively, a high apriori polarization score means that much of the polarization in the dataset is likely not driven by any single group.

$$Pol_{apr}(d) = \frac{1}{|d|} \sum_{c \in d} Pol_{apr}(c) \tag{6}$$

We compare this quantity with the average observed polarization of all comments:

$$Pol_{obs}(d, g) = \frac{1}{|d|} \sum_{c \in d} nDFU(A(c \mid g)) \tag{7}$$

We can now compare the polarization that is attributed to a single annotator group (Eq. 7) to a prior one (Eq. 6). We define our normalized metric, *apunim*, as:

$$apunim(d, g) = \frac{Pol_{apr}(d) - Pol_{obs}(d, g)}{1 - Pol_{apr}(d)} \tag{8}$$

Definition 3 (Apunim)**.** The degree to which an annotator group contributes to overall polarization in a set of comments.

- $apunim(g) \approx 0$: The examined group does not meaningfully disagree with the other annotators.
- $apunim(g) > 0$: The examined group has fundamental disagreements with the other annotators.
- $apunim(g) < 0$: The annotators of that group are polarized between themselves.

5.3 Statistical significance

Since annotation histograms may be noisy, we additionally assess whether the observed apunim value could arise by chance. To assess statistical significance, we construct a null distribution of apunim values from the randomized partitions and compare the observed apunim against this distribution using a one-sample Student-t test. Thus, the null hypothesis states that the observed polarization is indistinguishable from polarization induced by random annotator assignments:

$$\begin{aligned} H_0 &: \ \bar{apunim}_{\text{rand}}(d, g) = apunim(d, g) \\ H_a &: \ \bar{apunim}_{\text{rand}}(d, g) \neq apunim(d, g) \end{aligned} \tag{9}$$

where $\bar{apunim}_{\text{rand}}(d, g)$ denotes the average apunim values obtained from t randomized annotator partitions. For a detailed explanation of the p-value computation and the assumptions underlying the test, refer to App. C.3.

Group	Apunim	Support
DICES-350		
Ethnicity=Asian	-.066	8138
Education=Other	.021	2191
DICES-990		
Ethnicity=AfricanAm	-.091	5810
Ethnicity=Latino	.133	6610
Ethnicity=Other	-.074	24321
Ethnicity=White	.120	11392
Age=GenZ	-.063	13741
Age=GenX+	.046	14384
Education=High school	-.075	7419
Gender=Man	-.027	32494
Gender=Woman	.024	33471
Sap		
Ethnicity=Black	.042	4
Gender=Man	-.091	1439
Gender=Woman	.244	877
Kumar		
Innate		
Ethnicity=White	-.029	6619
Education=College	-.034	2508
SexOrient=Heterosexual	-.050	5721
IsTrans=No	-.079	1697
Experiences/Ideologies		
HasBeenTargeted=False	-.048	6632
IsParent=No	-.069	5514
IsParent=Yes	.049	6216
ReligionImportant=No	-.062	3864
ReligionImportant=Very	.052	4062
SeenToxicity=False	.035	3050
SeenToxicity=True	-.077	6360

Table 2: Statistically significant ($p < 0.05$) apunim results across the four datasets. Gray rows show `Ethnicity` and `Gender` groups. Consult App. I for the full results. "Support" refers to number of annotations.

6 Attributing real-life polarization to human groups

6.1 Experimental Setup

Assuming a confidence level of $\alpha = 0.05$ we test whether any of the provided PC dimensions can partially explain polarization in the annotations. The statistically significant results are presented in Table 2. Note that some groups are not included because there were no valid comments (see App. C.2). See App. E for details on other parameters and execution times, and App. I for the full results.

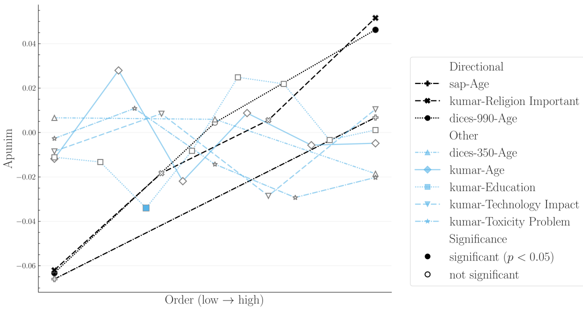

Figure 5: Polarization attribution trends in ordinal groups. The left side of the x-axis refers to young annotators, low education, low religiousness, and negativity towards the impacts of toxicity and technology. The full ordinal modes for each dimension and their individual labels can be found in App. I. We exclude groups with < 50 annotations, since they near-universally result in zero apunim values due to lack of data.

6.2 Which factors contribute to polarization?

Ethnicity Polarization is consistently attributed to ethnicity across all datasets, confirming prior work suggesting that it is a systematic source of annotator disagreement in these tasks (Goyal et al., 2022).

Gender Men and women have statistically significant, consistently opposite effects both in DICES-990 and Sap, consistent with previous findings regarding HS (Wojatzki et al., 2018) and toxicity annotation (Binns et al., 2017). Non-*transgender* annotators also showed the highest degree of polarization in the Kumar dataset in absolute terms ($-.788$) suggesting that transgender annotators are a systematic source of annotator disagreement, although only the inverse effect is detectable due to annotator sparsity (hence the negative sign).

Finding 3. *Annotator ethnicity and gender are consistent causes of polarization.*

Ideology appears to be a more consistent source of annotator disagreement than inherent person characteristics (e.g., personality, PC). Specifically, the Kumar dataset, which tracks ideological attributes, shows polarization attribution linked primarily to personal experiences and beliefs, rather than to sociodemographic dimensions. While the Kumar dataset is the sole source reporting these ideological attributes, further datasets are required to generalize this finding. The divide between ideology/personal experiences and innate sociodemographic characteristics has not been adequately explored in current literature and should be further studied in future work.

Religion demonstrates a clear relationship with polarization attribution: as annotators report higher levels of devoutness, their agreement increases. Fig. 5 illustrates that as annotators report greater belief, their consensus within the group improves.[11][12]

Finding 4. *Pol. attribution may be more pronounced in ideological attributes. It also escalates as annotators become more religious.*

7 The ideal subjective dataset

We briefly discuss the type of data most suited for our proposed methodology, and connect it with current NLP datasets.

How many annotators? Our methodology does not directly require numerous annotators overall, only a sufficient number per item—indeed, statistically significant results can be obtained with as little as four annotations (see Sap-AA; Table 11). A large annotator pool however is required to get sufficient coverage on the examined groups. More annotators allow both higher fidelity of each PC dimension, and more such dimensions.

How many annotators per item? We investigate observed polarization (since pol_{apr} relies on the empirical distribution) against varying numbers of annotators. Specifically, for every comment, we repeatedly sampled n annotators with replacement, calculated pol_{obs} for each sample and measured SD, averaging it across 1000 same-sized samples. As shown in Fig. 6, pol_{obs} exhibits a clear, inverse relationship with the number of annotators, attributed to better histogram estimation provided by nDFU (§5.1), though the rate of gain substantially diminishes after 20 annotators. Interestingly, some variability persists even with a very high annotator count.

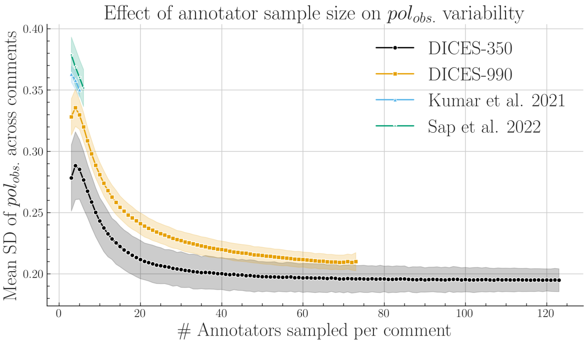

Figure 6: SD of pol_{obs} across comments, averaged across 1000 runs (Eq. 7). Shaded areas represent two SDs. Pol. estimates benefit from more annotators up to $n = 20$, at which point they begin to plateau.

How many items? Our methodology is designed to work with smaller datasets common in NLP. Three of the four datasets in the study feature low (300–600) item counts, while the Kumar dataset has been subsampled to 1% of its original size. Large datasets should be used only to provide larger coverage on the target domain, since a much better representation of individual groups can be achieved with more annotators per item, rather than more items.

8 Conclusion

We investigated to what extent polarization can be attributed to annotator groups. We discovered as of yet unknown issues regarding the presence of inherent and conflicting polarization, which we treat using a new metric, *apunim*. By applying this metric to four subjective NLP datasets, we discover that (1) ethnicity and gender seem to generally drive polarization attribution, and (2) ideological groups may be more important than reported in the literature. Finally, we empirically investigate and discuss how our methodology interacts with (generally small in both annotations and size) current subjective NLP datasets and release a PyPi library implementation to support further work.

Limitations

Researchers often release only aggregated single-label annotations or omit annotator PC

[11] The statistical insignificance observed during the transition from irreligious to highly religious groups may represent genuine shifts rather than inconclusive results, since Eq. 9 checks whether $apunim \neq 0$, given that $\overline{apunim}_{\text{rand}}(d, g)$ will usually be close to zero.

[12] This phenomenon may be also apply to other ordinal PCs, such as `Age` in both Sap and DICES-990, but finer-grained subgroups are needed to verify this.

characteristics entirely. As a result, both our analysis and the applicability of our framework are limited by the scarcity of suitable datasets. In addition, because datasets with large numbers of annotators per item are rare, our approach operates only at the dataset level. This limitation prevents us from identifying individual comments or examples that could serve as concrete explanations for the observed *apunim* values.

Additionally, our metric features some important practical limitations. It is computationally expensive when applied to very large datasets with a very large amount of PC dimensions each featuring multiple distinct subgroups (App. E). It also tends to underestimate polarization attribution when applied to many tens of thousands of items (App. H). Furthermore, the p-value estimates may falsely give the impression that zero-attribution effects are genuine, which should be taken into account when interpreting results. These limitations do not render our method insufficient—the first two issues can be solved by subsampling. We nevertheless believe that NLP datasets should generally move away from huge annotated corpora, and instead towards smaller datasets with large annotator groups per item that sufficiently represent social subgroups.

Finally, although we compare findings across datasets and relate them to prior work, we do not directly compare polarization and agreement metrics, since they rely on fundamentally different assumptions and mechanisms (see §2.2). These differences also make it difficult to fully explain certain patterns in our results, such as why some groups show statistically significant effects while others do not.

Ethical Considerations

All datasets used in this study were either publicly accessible, or were provided to us by the authors with full disclosure on the purpose of our research. For this reason, we do not release the data provided to us by the authors. No identifiable, personal information is included in our pipeline.

Our study uses examples of comments which may be considered divisive across certain demographics. These examples were created by us for demonstration purposes—their contents and the groups studied can be replaced with any suitable alternative. We do not claim that these examples necessarily reflect the views or behaviors of these groups, nor are they representative of our own opinions.

The purpose of our study is identifying systematic sources of polarization in subjective NLP datasets, in order to build more robust and equitable AI systems. However, as our methodology allows for the quantitative attribution of judgments to specific sociodemographic and ideological characteristics, we acknowledge three primary risks. First, the risk of discriminatory profiling: if the metric is applied outside of dataset quality analysis, its ability to correlate specific judgment patterns with annotator groups could be misused for surveillance or unjust profiling. Second, there is the risk of automated bias amplification: the diagnostic patterns of polarization we detect are reflections of human systemic biases. If subsequent models are trained on these patterns without critical human oversight, the AI risks automating and entrenching societal prejudices. Finally, the structural knowledge provided by *apunim* could be weaponized, allowing malicious actors to strategically target weak points in moderation systems or intentionally maximize societal conflict.

We stress that our metric serves as a diagnostic warning signal, requiring that any implementation in a deployed system must mandate a rigorous human-in-the-loop review, ensuring that the identified patterns lead to systemic change (e.g., policy reform, demographic diversity improvements) rather than merely justifying the automated system. We also actively encourage the scientific community to "red team" our methodology, helping to proactively identify and neutralize potential adversarial or discriminatory applications.

We have used small, open source LLMs (primarily LLaMa-8b and Qwen-7B) to help in small style changes throughout this document. We also used them in conjunction with a GPT-5 model to generate some of the code for the plots and experiments presented in this paper. All such outputs have been checked by the authors.

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A Acronyms

InPol Inherent Polarization

nDFU normalized Distance From Unimodality

FWER Family-Wise Error Rate

NLP Natural Language Processing

GPLv3 GNU General Public Licence v3

AA African American

PC Personal Characteristic

HS Hate Speech

SD Standard Deviation

B Annotation details

The number of annotators per comment for all datasets presented in §3 can be found in 8, and descriptive statistics can be found in Table 3. The DICES-990 dataset has exactly 123 annotations per comment, while the other datasets exhibit a few variations around the central mean reported in Table 1. Interestingly, the sample used from the Kumar dataset exhibits a few comments that vastly exceed the promised 5 number of annotators. We included these comments for the experiment presented in Table 2, but not for the annotator number experiment in 6 since they are very likely a data error. Tables 4,5,6,7,8 show the annotation counts for each PC group in each dataset.

C Statistical considerations

C.1 Defining apriori polarization

In §4.1, we explained how our methodology needs to compare the polarization of $A(c)$ and $A(c \mid g)$. Direct comparisons are not advisable, since we would be comparing statistics between a set and its subset—in this case, standard practice dictates comparing the subset with its complement; i.e., the in-group vs. the out-group instead of the in-group vs. all the annotations. However, this also would not work; if we apply the subset-vs-complement method in the example of Fig. 2, we would end up subtracting the nDFUs of men and women, which are similar as they are both unimodal distributions, falsely concluding that there is no polarization present.

C.2 Filtering ineligible comments

Because *apunim* is fundamentally designed to quantify polarization, we exclude comments that exhibit no measurable polarization

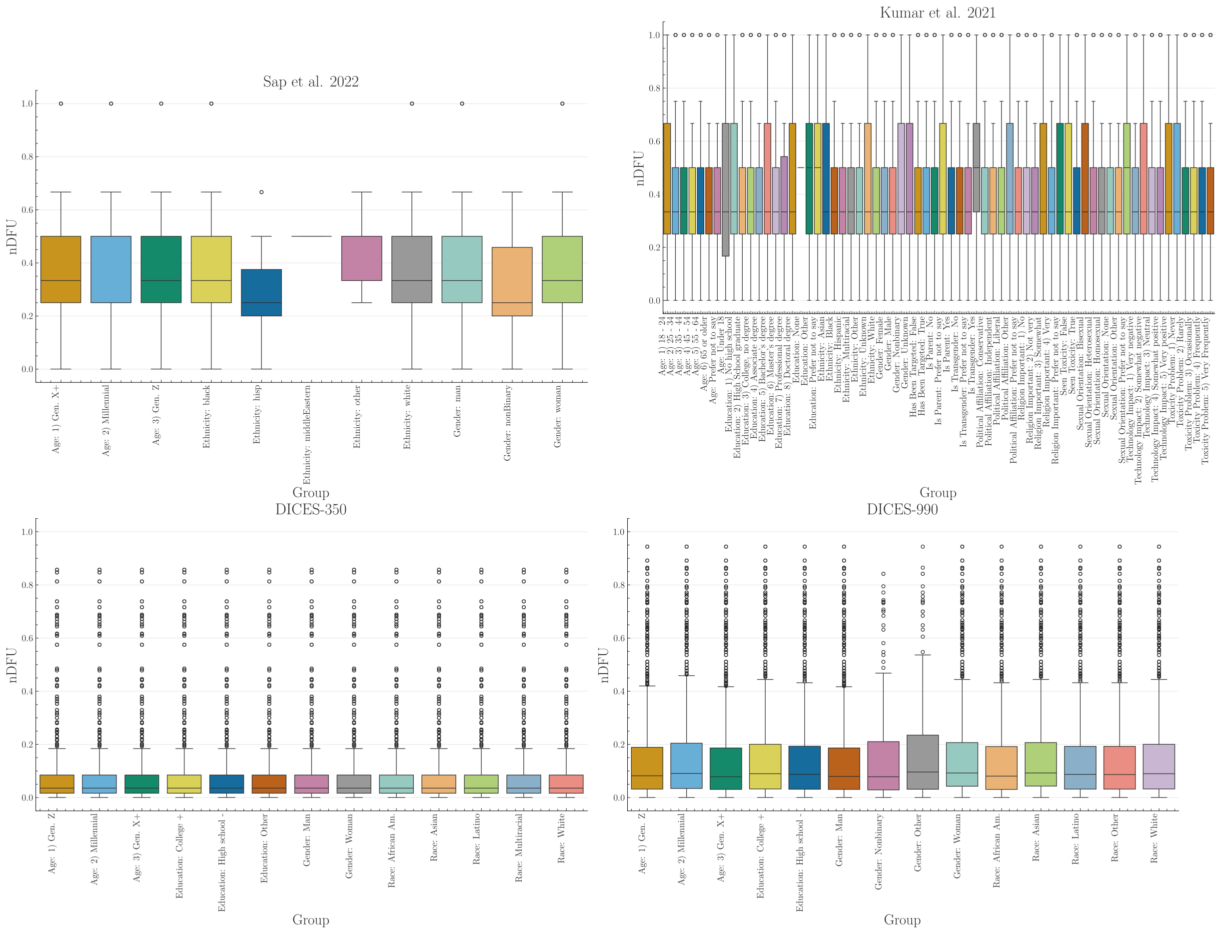

Figure 7: Histogram of dataset polarization for the human datasets considered in this paper per PC dimension.

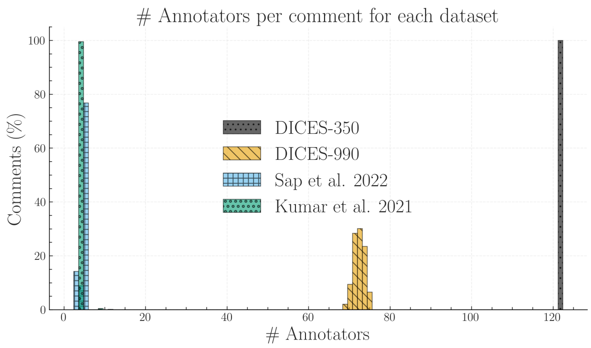

Figure 8: We calculate the ratio of comments for each dataset (y-axis) that had x number of annotators (x-axis). We truncate the x-axis to 130 annotators, disregarding very few comments in the Kumar dataset exceeding this threshold (see Table 3), which is likely caused by a data error.

($nDFU(c) = 0$). Furthermore, we also exclude comments annotated exclusively by a single group (i.e., comments where there are no two groups with at least three annotators each), since we require valid histograms for the non-parametric estimation of nDFU (see Eq. 1). This is an unfortunately common scenario, since many NLP datasets feature only $5-6$ annotations per comment. In a best-case scenario of 5 annotators split evenly between two groups, we would get a 3-2 split.

C.3 P-value estimation

Since our test is parametric (§5.3), it assumes that there exist (1) enough data-points to reliably run the means-test, (2) enough annotations per PC group to estimate nDFUs in Equations 5-7, (3) the apunims of the random partitions are independent and identically distributed. Condition (1) can be safely assumed for most datasets, because they feature enough items to reliably assume normal residuals (see §3). We discuss condition (2) in §7. Condition (3) is partly fulfilled; the partitions are identically distributed, but may be dependent on the sample distribution.

Replacing the parametric means test with

Dataset	count	mean	SD	min	25%	50%	75%	max
DICES-350	350	123	0	123	123	123	123	123
DICES-990	990	72.8	1.17	69	72	73	74	76
Sap	585	5.6	.78	3	6	6	6	12
Kumar	2998	5.02	.34	5	5	5	5	10

Table 3: Number of annotations per item in each of the four datasets used in this study.

Category	Subgroup	Support
Gender	Woman	21700
	Man	21350
Race	White	10500
	African Am.	10150
	Asian	9100
	Latino	7700
	Multiracial	5600
Age	1) Gen. Z	19600
	2) Millennial	12600
	3) Gen. X+	10850
Education	College +	26250
	High school -	14350
	Other	2450

Table 4: Subgroup Counts for DICES-350

Category	Subgroup	Support
Gender	Woman	36050
	Man	35392
	Nonbinary	331
	Other	330
Race	Other	26394
	Asian	19885
	White	12322
	Latino	7170
	African Am.	6332
Age	Gen. Z	14899
	Millennial	41542
	Gen. X+	15662
Education	College +	64054
	High school -	8049

Table 5: Subgroup Counts for DICES-990

a non-parametric alternative would relax assumptions about the underlying distribution but would not resolve the potential dependence between the random apunims. We initially attempted to implement a permutation test on the empirical (random apunim) distribution, but this resulted in excessive computational costs.

While the test operates on a single group in principle, in practice we often want to test polarization for all subgroups of a specific dimension. Since we are simultaneously considering multiple hypotheses, we apply a multiple comparison correction to the resulting p-values—specifically the Holms method (Holm, 1979). The test is parameterized by the Family-Wise Error Rate (FWER), which is used to tune the strength of the correction; we can increase this value to make our test more conservative towards multiple hypotheses (Chen et al., 2017)). In general, it is safe to set FWER equal to the significance level of our test (e.g., $FWER = 0.95$ if we are looking for $p < 0.05$).

D Software

D.1 Installation and usage

We release the `apunim` package, which is a Python library that implements both the nDFU and apunim calculations, as well as its statistical test. The software is available on PyPi We also provide high-level documentation in HTML format that includes an introduction, guide, and full low-level documentation for each of the two provided functions (`https://anonymous.4open.science/r/apunim-page`).

The code is open-source, licensed under the GNU General Public Licence v3 (GPLv3) and can be found on the project's repository (`https://anonymous.4open.science/r/aposteriori-unimodality-E8C1`). The library is cross-platform, works for any Python 3 version, and requires very few dependencies (namely `numpy, scipy, statsmodels`).

Category	Subgroup	Support
Age	Gen. Z	252
	Millennial	2259
	Gen. X+	786
Ethnicity	white	2168
	black	1091
	hisp	32
	other	5
	middleEastern	1
Gender	man	1859
	woman	1412
	nonBinary	26

Table 6: Subgroup Counts for Sap et al. 2022

D.2 Implementation details

By default, the computation of the apunim values happens on a single thread. In cases of serious computational cost, we recommend running the apunim calculation on multiple concurrent processes, since all operations are CPU-bound and thread-safe. This may be useful in cases where there are multiple PC dimensions, each of which is computationally expensive.

In order to lower the computational cost required, we re-use the random partitions $\tilde{r}_i$), which are originally calculated to obtain the apriori polarization values (see Eq. 5), for the p-value calculations (§5.3). This means that in our current implementation, the number of groups for estimating apriori polarization ($\tilde{r}_i$ in Eq. 5), and the number of apunim values used in the p-value calculation have to be the same.

E Replication Details

We use version `1.0.2` of the `apunim` library for our experiments. We use 100 random iterations for the apriori value calculation (Eq. 5) and the p-value estimation (§5.3), and set a FWER rate of .95.

While the library efficiently handles large numbers of annotators through vectorized operations, performance degrades significantly when processing a high volume of comments, particularly when annotators belong to multiple distinct groups. For instance, while experiments across all datasets in §3 complete within a minute, the Kumar dataset necessitates approximately one hour of computation using commercial hardware. Furthermore, the annotator variance analysis detailed in §7 required over ten hours. While this latter time requirement is anticipated due to the inherently demanding nature and non-standard application of our library, it demonstrates the existence of a potential scaling bottleneck. The same applies to a lesser extent for the subsampling experiments on the DICES datasets, taking collectively three hours of computation.

Gender	
Female	8011
Male	6832
Unknown	124
Nonbinary	93
Ethnicity	
White	10699
Black	1940
Asian	862
Multiracial	664
Hispanic	418
Other	247
Unknown	230
Age	
Under 18	3
18 - 24	1711
25 - 34	5969
35 - 44	3785
45 - 54	1977
55 - 64	1101
65 or older	471
N/A	43
Education	
No high school	71
High School graduate	1353
College, no degree	3090
Associate degree	1676
Bachelor's degree	6071
Master's degree	2261
Professional degree	236
Doctoral degree	198
N/A	71
Other	32
Sexual Orientation	
Heterosexual	12391
Bisexual	1665
Homosexual	515
N/A	302
Other	131

Table 7: Subgroup Counts for the Kumar et al. 2021 dataset. Part 1 of 2.

We hypothesize that this bottleneck is likely caused by repeated creation of numpy arrays, resulting in repeated memory allocations for each subgroup, in each comment, for each PC dimension. We plan on profiling and addressing it in the next releases of our software. Until then, we suggest either using multiprocessing for each PC subgroup, or running the different experiments in parallel (indeed, we use the `gnu-parallel` package (Tange, 2025) for our own experiments). We also note that the Kumar dataset also requires substantial RAM resources, estimated around 42–46GB of dedicated RAM should the whole dataset be considered.

Finally, we note that in Fig. 6, we removed values where there was a lack of items annotated by the specified number of annotators (e.g., only 7% of comments in DICES-990 were annotated by more than 73 annotators—see App. B).

F Synthetic experiments

This appendix details the synthetic experiments used to illustrate the functionality and behavior of nDFU. In all simulations, the bin size is set to $N_{bins} = 10$.

Disagreement vs. polarization To explore the relationship between annotation polarization and disagreement in Fig. 1, we constructed a 2×2 matrix of synthetic datasets, each drawing from a pool of $N_{annotators} = 100$. Polarization is simulated by defining a minority group, which constitutes 20% of the total annotators (n_special = 20). These minority annotators are assigned a targeted mean shift (mean_shift = 1) from the base distribution. The base variance for these distributions is set at variance = 0.3. Different types of distributions (unimodal, bimodal, multimodal, uniform) are achieved by controlling the base_means and applying variance multipliers (e.g., variance $\times$ 3 for multimodal).

- The top-left panel ("Minority split") shows high polarization and low disagreement.
- The top-right panel ("Mixed opinions") shows both high polarization and high disagreement.
- The bottom-left panel ("No disagreements") shows low polarization and low disagreement (consensus).
- The bottom-right panel ("Overall mixed opinions") shows low polarization but high disagreement.

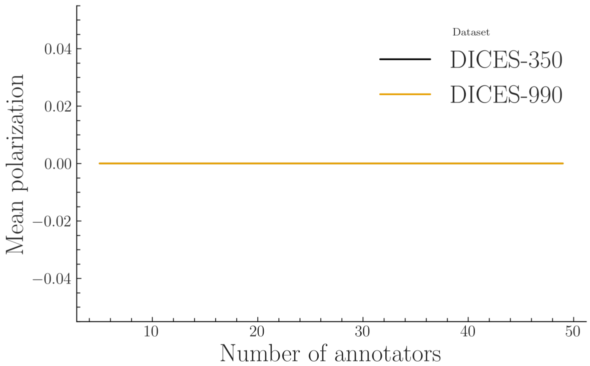

Figure 9: InPol for both DICES datasets averaged across ten runs, subsampling annotators starting from five to the actual number of annotators with a step of two. InPol remains zero, no matter how many annotators per comment are considered.

Single-comment polarization We simulate the distribution of toxicity annotations in Fig. 2 across two demographic groups (Men and Women). The distribution of Men's scores is modeled using a truncated normal distribution with loc = 2 and scale = 1.3, while Women's scores are modeled with loc = 8 and scale = 1.3. The combined dataset, representing all annotations, is used to demonstrate the impact of population weighting on the overall nDFU score. The sample size for each group in this visualization is $N_{annotators} = 50$.

Discussion-level polarization We examine nDFU scores for three specific textual examples presented in Fig. 4. For the individual comments, $N_{annotators} = 200$ for each group. In the first case (misogynist comment), Men's scores are modeled with loc = 6, while Women's scores are modeled with loc = 2. In the second case (misandrist comment), the roles are reversed. The third visualization combines distributions for men and women derived from the respective single-comment distributions.

G How does Inh. Pol. look like?

Table 9 shows a random sample of comments taken from the Sap dataset and their respective InPol values (§4.2). Note that explicitly toxic or racist comments tend to have InPol values of zero, alongside clearly unproblematic comments. Inversely, more ambiguous and less comprehensible comments tend to have larger

InPol values. This pattern may suggest that InPol may be tied to the overall ambiguity of the statements, especially in cases where comments may either be interpreted as very toxic, or non-toxic, which agrees with the overall goal of nDFU (§2.2). We include the InPol values for each individual comment for all four datasets in the supplementary material.

In order to investigate whether the numerous annotators per comments can explain the lack of InPol in the DICES datasets (§4.2.4) we gradually subsample the annotators for each comment randomly with replacement, starting from five annotators per comment, and scaling up to the full complement of annotators in the dataset with a step of two. For each such step, we choose ten annotator samples and compute the mean InPol across all runs. Fig. 9 demonstrates that even with annotator counts similar to the Kumar and DICES datasets, InPol remains zero, suggesting that the absence of InPol can not be attributed to annotator number alone—rather, it must be a pattern attributable to the DICES datasets themselves.

H Ablation experiments

Since our approach was designed for the small-scale datasets common in NLP, we were cautious when testing it on the massive Kumar dataset. Because of the data volume, we analyzed various smaller samples (1,000, 10,000, and 30,000 comments), shown in Table 10. As sample size increased, the statistical metrics (apunim values) consistently shrank, and we found no statistically significant effects in the larger samples. This outcome leads to an important finding: the statistical test becomes less effective when applied to very large datasets.

The test is overly conservative, meaning it almost always suggests there is no significant effect, even when an effect likely exists in reality. This inability to detect a real signal is a Type II Error (a false negative). This suggests that our test is statistically weak or "underpowered" in massive settings. Since our methodology is built for the small-scale NLP datasets that dominate the research field, we deem this limitation acceptable for the scope of our work.

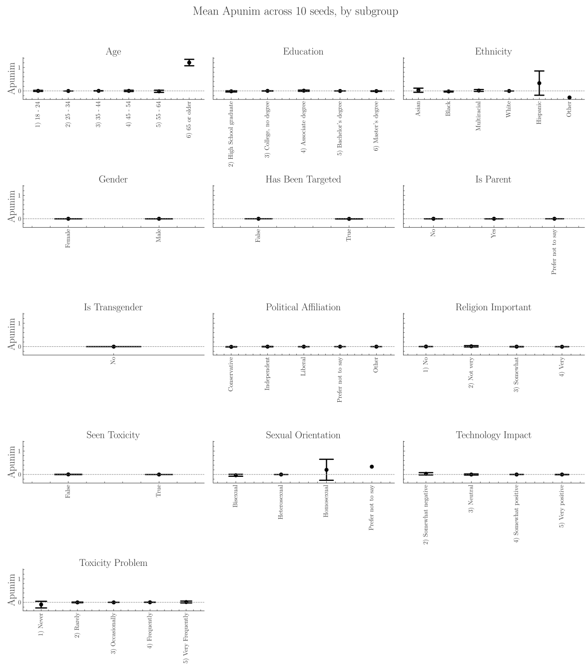

Figure 10: Apunim values on the Kumar dataset ($n_{samples} = 3000$) across ten runs with different subsets. Black bars indicate two SDs. PC dimensions with many groups are more volatile, although all dimensions broadly keep the same apunim sign across subsets.

Astute readers may notice that some variability does exist across differently sized samples for the same examined groups, where for example the sign of the apunim value may switch. In order to verify that this variability is caused by the sample size, and not by different samples being drawn, we execute a further ablation study. Specifically, we execute the apunim study on the 3000-sized sample of the dataset, using different subsets of the Kumar dataset.

The results can be found in Fig. 10, where we note that variability is not an issue, except for a couple of cases. These cases seem to represent sparse annotator groups, where missing a few comments may genuinely change the value of the apunim metric. These groups however are largely not detected by apunim as significant either way (see §6).

I Full results

Is Transgender	
No	14549
Yes	363
N/A	148
Political Affiliation	
Liberal	5997
Independent	4143
Conservative	3981
N/A	609
Other	330
Is Parent	
Yes	7889
No	7015
N/A	156
Technology Impact	
Very negative	149
Somewhat negative	1411
Neutral	1905
Somewhat positive	7690
Very positive	3905
Toxicity Problem	
Never	869
Rarely	2485
Occasionally	5307
Frequently	4441
Very Frequently	1958
Religion Important	
No	4735
Not very	1808
Somewhat	3548
Very	4741
N/A	228
Seen Toxicity	
True	11505
False	3555
Has Been Targeted	
False	10741
True	4319

Table 8: Subgroup Counts for Kumar et al. 2021 dataset. Part 2 of 2.

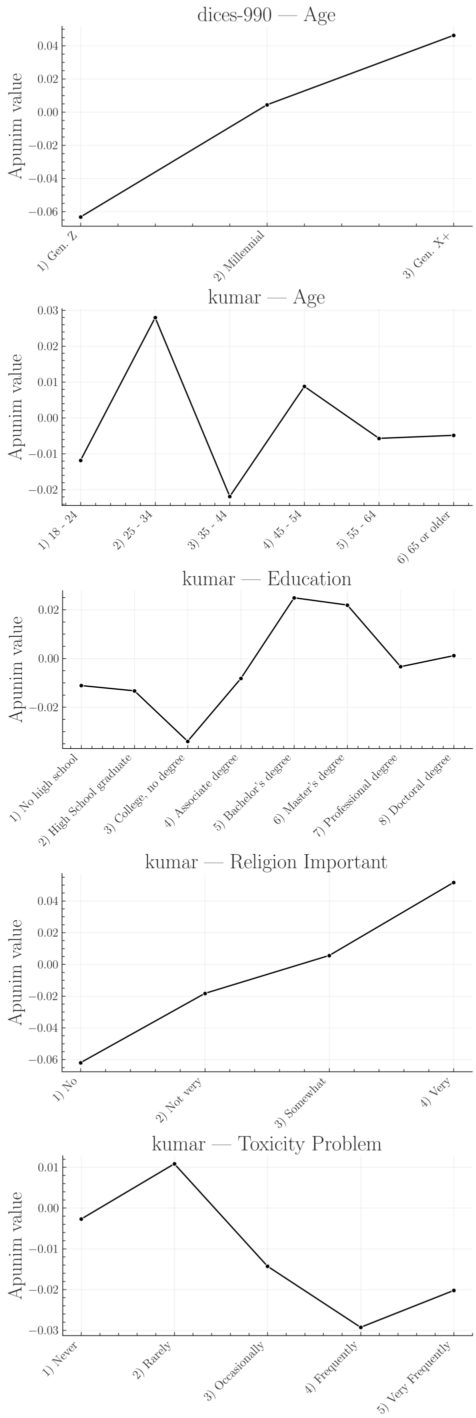

Figure 11: Apunim values expanded for each ordinal PC dimension. This figure is complementary to Fig. 5. The values in the graphs were extracted from the Tables of App. I.

Comment	InPol
The minks mentioned in the headline are indeed ”rising” from where they were based, but that is due to them being buried in shallow mass graves (Africa) and filling with gas that causes them to rise out. Rest assured, they’re still very much dead, thankfully!.	1
There should be such thing as a bitch resistance for us women	1
There’s a big difference between being a bitch and being a woman	1
i just watched a video of @joisnotonfire being an annoying hoe on skype i’m laughing	1
niggas love to commit crime	1
I swear to God above I love ya and when ya be with this brotha, I swear I love you my bro as long as you not a nigga	0.5
I will become a cold ass bitch because of my treatment	0.5
I would happily lay out a nigga	0.5
Shoot yourself in the head you hypersensitive fuckin bitch.	0.5
not all women are bitches, but all bitches are women	0.5
“ you ugly bitch!!!”- that seems to be my real name these days”	0
“I am happy to report that drowning does kill this demonic entity. Oh holy are the waters of Chernarus!”” Omg, DED ;)”	0
“Let them praise the name of the LORD: for his name alone is excellent; his glory is above the earth and heaven.”” Psalms 148:13 #Bible”	0
someone plz get this nigger some help. He’s playing up like mad	0
stop talking you fucking nigger !	0

Table 9: Sample of comments from the Sap dataset and their InPol. Note that unambiguously toxic comments can be assigned with zero InPol, while more nuanced cases tend to have higher values.

PC	1000	10000	30000
Age			
8 - 24	-.00235	-.03667	-.01614
25 - 34	.04544	.00036	.00045
35 - 44	-.04315	-.00070	-.00070
45 - 54	.00144	.01310	.00305
55 - 64	.00215	-.05660	.00113
65 or older	-.00657	N/A	.94805
Education			
High School graduate	-.02302	-.00531	.01411
College, no degree	-.04546	.00952	.00590
Associate degree	.00106	.00539	.00888
Bachelor's degree	.02875	-.00104	-.00029
Master's degree	.02418	.00374	-.00027
Ethnicity			
Asian	-.01177	.01076	.00664
Black	.03590	-.00417	-.01162
Hispanic	.00379	.0	-.09910
Multiracial	.00755	.0	.0
Other	-.00813	N/A	-.30556
White	-.02301	-.00014	.00013
Gender			
Female	-.00620	.00165	.00053
Male	.04037	-.00234	-.00132
Has Been Targeted			
False	-.05945	-6.35e-05	1.44e-05
True	.01669	-.00345	-.00200
Is Parent			
No	-.08405	.00077	.00075
N/A	.00442	1.0	3.89e-16
Yes	.07057	-1.02e-05	.00015
Is Transgender			
No	-.10182	.0	2.24e-16
Political Affiliation			
Conservative	.05553	-.00266	-.00239
Independent	-.00321	.00101	.00237
Liberal	-.02626	.00095	.00070
Other	-.00324	.0	.0
Religion Important			
No	-.07148	.00232	.00140
Not very	-.01344	.01005	.00567
Somewhat	.02157	.00314	-.00220
Very	.04283	-.00304	-.00165
Seen Toxicity			
False	.02454	-.00144	-.00075
True	-.08237	.00064	.00048
Sexual Orientation			
Bisexual	-.00272	-.03953	-.01297
Heterosexual	-.06722	-.00032	1.87e-06
Homosexual	-.01172	.0	.05263
Technology Impact			
Somewhat negative	-.00807	.0	.00258
Neutral	.01699	-.01362	.00030
Somewhat positive	-.05332	-.00012	.00074
Very positive	.02712	-.00421	-.00303
Toxicity Problem			
Never	-.00769	-.03661	-.08076
Rarely	.00410	.00322	-.00113
Occasionally	-.01189	-.00261	-.00118
Frequently	-.03226	-.00138	-.00012
Very Frequently	-.01684	.02953	.01461

Table 10: Comparison of apunim values across different sample sizes (1k, 10k, 30k) for the Kumar dataset. Rows with more than one N/A value removed.

Group	apunim	pvalue	support
	Kumar		
Age=18-24	-.0118	.4822	1431
Age=25-34	.0279	.1185	4888
Age=35-44	-.0219	.1733	3102
Age=45-54	.0088	.5394	1668
Age=55-64	-.0057	.6592	938
Age=65 or older	-.0049	.6747	409
Age=N/A	.0000	N/A	37
Age=Under 18	.0000	N/A	2
Ed=No HS	-.0111	.5278	60
Ed=High School	-.0133	.3461	1123
Ed=College	-.0340	.0189	2508
Ed=Associate	-.0082	.5963	1362
Ed=Bachelor's	.0249	.1091	5007
Ed=Master's	.0219	.1091	1982
Ed=Professional	-.0034	.7459	192
Ed=Doctoral	.0011	.8651	164
Ed=Other	.0000	N/A	28
Ed=N/A	.0000	N/A	63
Eth=Asian	-.0019	.9332	720
Eth=Black	.0143	.7070	1671
Eth=Hispanic	-.0075	.9092	352
Eth=Multiracial	-.0026	.9332	540
Eth=Other	-.0007	.9332	204
Eth=Unknown	-.0063	.9092	189
Eth=White	-.0291	.0416	6619
Gen=Female	-.0245	.0720	6141
Gen=Male	.0261	.0720	5479
Gen=Nonbinary	-.0014	.9436	81
Gen=Unknown	-.0008	.9436	104
Targeted=False	-.0482	1.04e-05	6632
Targeted=True	.0146	.2310	3643
IsParent=No	-.0686	2.39e-08	5514
IsParent=N/A	.0017	.8913	130
IsParent=Yes	.0490	2.85e-05	6216
IsTrans=No	-.0788	7.47e-05	1697
IsTrans=N/A	-.0007	.9483	128
IsTrans=Yes	-.0073	.7711	340
Pol=Conservative	.0269	.0534	3386
Pol=Independent	-.0201	.1371	3432
Pol=Liberal	-.0278	.0534	4945
Pol=Other	-.0132	.2947	271
Pol=N/A	.0034	.6523	511
RelImp=No	-.0620	2.96e-07	3864
RelImp=Not very	-.0183	.1039	1493
Ed=Somewhat	.0056	.6169	3012
RelImp=Very	.0516	4.58e-05	4062
RelImp=N/A	.0066	.2968	194
SeenTox=False	.0350	.0029	3050
SeenTox=True	-.0766	4.82e-13	6360
SexOrient=Bisexual	.0087	.7884	1481
SexOrient=Heterosexual	-.0500	3.22e-05	5721
SexOrient=Homosexual	-.0048	.7884	430
SexOrient=Other	.0009	.9055	115
SexOrient=N/A	-.0061	.7884	265
TechImp=Very neg	.0000	N/A	118
TechImp=Somewhat neg	-.0085	.4097	1179
TechImp=Neutral	.0085	.4097	1603
TechImp=Somewhat	-.0285	.0691	6040
TechImp=Very pos	.0105	.4097	3315
ToxProblem=Never	-.0027	.8071	741
ToxProblem=Rarely	.0108	.3965	2087
ToxProblem=Occ	-.0143	.3809	4347
ToxProblem=Freq	-.0293	.0507	3740
ToxProblem=Very	-.0202	.1404	1625

Table 11: Apunim results for the Kumar dataset. * denotes $p < 0.10$, ** denotes $p < 0.05$, and *** denotes $p < 0.01$. Rows shaded in gray indicate a negative apunim value.

Group	apunim	pvalue	support
	Sap		
Age=GenX+	-.066	.708	271
Age=Millennial	.007	.744	1886
Ethnicity=black	.042	4.000	4
Ethnicity=white	-.000	.998	1896
Gender=Man	-.091	.000	1439
Gender=Woman	.244	.000	877

Table 12: Apunim results for the Sap Dataset. * denotes $p < 0.10$, ** denotes $p < 0.05$, and *** denotes $p < 0.01$. Rows shaded in gray indicate a negative apunim value. Groups for which no polarized comments among groups not included.

Group	apunim	pvalue	support
	DICES-350		
Age Gen. Z	.007	.539	17528
Age Millennial	.006	.539	11268
Age Gen. X+	-.019	.166	9703
Ed=College +	-.012	.234	23475
Ed=High school	-.015	.186	12833
Ed=Other	.021	.025	2191
Gender Man	.019	.058	19093
Gender Woman	-.021	.058	19406
Race African Am.	.043	2.41e-05	9077
Race Asian	-.066	1.16e-10	8138
Race Latino	.003	.920	6886
Race Multiracial	.000	.980	5008
Race White	.020	.067	9390

Table 13: Apunim results for the DICES-350 Dataset. * denotes $p < 0.10$, ** denotes $p < 0.05$, and *** denotes $p < 0.01$. Rows shaded in gray indicate a negative apunim value. Groups for which no polarized comments among groups not included.

Group	apunim	pvalue	support
	DICES-990		
Age Gen. Z	-.063	3.05e-25	13741
Age Millennial	.004	.495	38436
Age Gen. X+	.046	7.17e-15	14384
Ed=College +	-.009	.163	59142
Ed=High school	-.075	5.80e-42	7419
Gender Man	-.027	3.81e-05	32494
Gender Woman	.025	.00015	33471
Race African Am.	-.091	1.61e-64	5810
Race Asian	-.011	.080	18428
Race Latino	.133	7.57e-112	6610
Race Other	-.075	1.51e-30	24321
Race White	.120	6.03e-81	11392

Table 14: Apunim results for the DICES-990 Dataset. * denotes $p < 0.10$, ** denotes $p < 0.05$, and *** denotes $p < 0.01$. Rows shaded in gray indicate a negative apunim value. Groups for which no polarized comments among groups not included.